PHASE 1 USABILITY AUDIT

Weber State University Class Registration

A usability and accessibility audit of the Register for Classes app in the eWeber portal

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Course
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Date
October 9, 2026

Repository
github.com/Kaden-13/Contextual-Research

1. Executive Summary

This audit evaluated the Register for Classes app in Weber State University's eWeber portal, the tool students use to find, compare, and add classes each semester. Registration is high stakes: a mistake can cost a seat, a semester, or tuition money, yet the interface makes students work around it rather than with it.

The audit combined a contextual inquiry with 2 students (one new to registration, one experienced), an expert review against Nielsen's 10 Usability Heuristics, and a WCAG 2.1 AA accessibility audit using WAVE and manual testing.

The most serious findings:

1. **A keyboard only user cannot register.** Pressing Tab skips every listed class and every Add button (WCAG 2.1.1, severity 4).
2. **Clicking Add does not register you.** Add only places a class in a pending list, and nothing is registered until a separate Submit button in the far corner is clicked (H1, H5).
3. **Class times are hard to find, and prerequisites aren't on the page at all.** The novice participant asked "Where do i see the time for the class" and needed help, and both participants had to leave the page (CatTracks or an advisor) to check eligibility.
4. **Errors don't explain themselves.** Trying to add a full class produces a vague error that never says the class is full or what to do next (H9, WCAG 3.3.3).
5. **The page where students choose classes is the least accessible page in the flow.** Once search results load, WAVE errors jump from 3 to 23, contrast errors from 1 to 89, and the AIM Score drops from 8.7 to 2.5 out of 10.

In the sessions, the novice student needed help just to find class times, took more than twice as long as the experienced student, and rated the experience 1 out of 5. The experienced student moved quickly, but only because she normally avoids the portal, relying on the Schedule Builder and her advisor instead.

Overall accessibility result: The registration flow **fails** WCAG 2.1 AA.

Fix first: Make every Add button reachable and visible by keyboard, clearly separate "pending" from "registered," and show class times and prerequisites in plain text on each result. Those changes remove the biggest legal risk, the most costly student mistake, and the biggest source of confusion in the sessions.

> **Key numbers**
>
> - 14 heuristic issues, 6 rated major (severity 3)
> - 12 WCAG 2.1 AA findings, 1 rated severity 4
> - WAVE AIM Score on the search results page: **2.5 / 10**
> - Lowest measured contrast: **1.07:1** (requirement is 4.5:1)
> - 5 of 6 user tasks completed without help; ease ratings of 1 (novice) and 4 (experienced) out of 5

2. Problem Statement and Context of Use

Weber State University students register for classes through the eWeber portal, using either the Register for Classes (Add/Drop) app or the Registration Schedule Builder. This audit focuses on the Register for Classes app: the path a student follows to find a class, compare sections, confirm eligibility, and add a class to their schedule.

The primary users are Weber State undergraduates. Many are traditional students between 18 and 24, but a large share are working, commuting, or returning students who register in short sessions between jobs and other obligations, often on a phone. The system also needs to serve students who rely on screen readers, magnification, or keyboard only navigation.

Registration is time sensitive and high stakes. Seats in popular sections fill quickly, and a wrong choice can delay graduation. Weber State also does not drop students for nonpayment or nonattendance, so a student who believes they dropped or never registered for a class can still be charged. When the interface relies on registrar codes, spreads key information across the page, or explains errors in system terms, students lose time, make costly mistakes, or give up and turn to an advisor or friend. For the university, that means more load on advising and the registrar, slower progress toward degrees, and legal exposure under the ADA and Section 504 if the system is not accessible.

Core tasks evaluated

1. Find a class of the participant's choosing
2. Compare sections by time, delivery method, and open seats
3. Confirm eligibility and locate how to add a section
4. Recover from registration errors (evaluator review)

3. Methodology

This audit combined three methods so that findings from one could be checked against the others.

Contextual inquiry. I observed 2 Weber State students using their own devices and accounts: one who was new to the registration system (P1) and one who had registered many times (P2). Choosing opposite ends of experience made it possible to compare what blocks a first time user with what an experienced user has learned to work around. Each participant completed 3 realistic tasks while thinking aloud, then answered short follow up questions. Sessions were not audio or screen recorded. Instead, I took live notes on task completion, time on task, wrong turns, hesitations, workarounds, and word for word quotes, then cleaned them up the same day. So that no one's real schedule changed, sessions were a mock run: participants searched current semester classes and showed how they would add a section, but did not actually add or submit anything. The full script is in `02_user_research/interview_protocol.md`, and notes are in `02_user_research/raw_notes/`.

Synthesis. I broke the notes into individual observations and grouped them through affinity mapping into themes, giving the most weight to problems both participants hit or that the heuristic evaluation confirmed (`02_user_research/synthesis/affinity_themes.md`).

Heuristic evaluation. I walked through each screen of the flow against Nielsen's 10 Usability Heuristics, used browser DevTools to inspect buttons, styles, and network calls, and rated every issue from 0 (not a problem) to 4 (usability catastrophe) based on frequency, impact, and persistence (`03_heuristic_evaluation/heuristic_evaluation_log.md`).

Accessibility audit. I scanned 5 pages of the flow with the WAVE browser extension, then manually tested color contrast (SC 1.4.3), keyboard access and visible focus (SC 2.1.1 and 2.4.7), zoom at 200% and 400% (SC 1.4.4 and 1.4.10), and semantic structure and ARIA support (`04_accessibility_audit/wcag_audit_checklist.md`).

Limitations. With only 2 participants, both of whom I know personally, the research shows the kinds of problems students run into, not how often they occur across all students. Participants who know the researcher may also soften criticism, so I emphasized that the system was being tested, not them. The evaluation was done by a single evaluator. Because nothing was actually submitted, the final registration step, system behavior under heavy load, and real seat competition were not observed; the full class error is based on my own past registration experience. The experienced participant worked silently, so her data comes from her behavior and end of session answers rather than quotes, and Task 3 times were not recorded.

4. User Research Findings

Participants

ID	Experience	Year and major	Device
P1	Novice, registered once before	Freshman	iPhone, then switched to a computer
P2	Experienced, registered 5 times	Junior, Radiology	Laptop

Task Results

Task	P1 (novice)	P2 (experienced)
T1: Find a class	Completed alone, 15 min	Completed alone, 5 min
T2: Choose a section that fits a schedule	**Needed help**, 20 min, confidence 2/5	Completed alone, 10 min, confidence 5/5
T3: Check eligibility and find how to add	Completed alone	Completed alone
Ease rating (1 to 5)	**1**	4

5 of 6 tasks were completed without help. The novice spent 35 minutes on the first two tasks, compared with 15 for the experienced student.

> **In their words**
>
> - “Where do i see the time for the class” (P1)
> - “Why do i have to click so many buttons” (P1)
> - “I had to switch to the computer because it was just to much trying to do it on my phone” (P1)

Theme 1: “Where do I see the time?” Class details are hard to read

The novice couldn’t find meeting times on her own and needed help, the only assisted task in the study. Even the experienced student named this task the hardest because it “takes longer to find a class that works with my schedule and with other courses.”

Mental model mismatch: Students expect class times to be the most visible fact in a result. Instead, days appear as tiny gray letter boxes (S M T W T F S) with one highlighted, packed into a cell with Type and Building text that is often cut off. (Supports HE13, HE06, HE11.)

Theme 2: “Why do I have to click so many buttons?” Registration is spread across too many tools

Neither participant checked eligibility inside Register for Classes. The novice opened CatTracks in a separate tab and called going back and forth between them the hardest part of the session. The experienced student asks her advisor instead. The novice also couldn’t understand why one class takes so many steps, and said she would “merge CatTracks and the registration portal.”

Mental model mismatch: Students expect one place to find a class, see whether they can take it, and register. The system splits that across Register for Classes, CatTracks, and advisors. (Supports HE14, HE10.)

Theme 3: Add isn’t registering, and students learn that the hard way

The first time the novice registered, she thought clicking Add was all she had to do, and only realized otherwise when she spotted the Submit button. She also hesitated at choosing the “Enrolled” option before submitting. The experienced student knows Add “adds to schedule but doesn’t register” and checks the bottom right corner to confirm. That rule is learned through experience, not explained by the interface. (Confirms HE01, HE04, HE08.)

Theme 4: Experienced students route around the system

The experienced student finished every task quickly and silently, but said she doesn’t normally use the registration portal at all. She relies on the Schedule Builder and her advisor, and the one thing she’d change is having to use the portal. Her smooth session reflects workarounds, not a smooth system, and shifts work onto advisors.

Theme 5: “Too much on my phone”

The novice started on her iPhone and switched to a computer because the page was too much on a small screen. This matches the zoom and reflow failures in the accessibility audit. (Supports HE12, A05.)

Novice vs. Experienced

	P1 (novice)	P2 (experienced)
Needed help	Yes, finding class times	No
Understands Add vs. Submit	Learned after a mistake	Yes
Where prerequisites come from	CatTracks, in a separate tab	Advisor
Uses the portal normally	Yes	No, avoids it
Ease rating	1	4

5. Heuristic Evaluation Findings

The Mental Model Gap

Students approach registration like an online shopping cart or seat reservation: find the class, click Add, and it’s yours. The system works differently, and most major issues come from that gap.

Students expect	What the system actually does
Clicking Add reserves my seat.	Add only places the class in the Summary as pending. Nothing is registered until Submit.
I can quickly see if a section has room.	Seat counts are shown, but every section also shows a waitlist line, even when seats are open.
If a class is full, the system tells me and shows my options.	A vague error appears that doesn’t say the class is full.
Dropping a class is a visible button.	Drops are hidden in an Action dropdown, then confirmed with Submit.
Not attending or not paying cancels a class.	Nothing is dropped automatically; the student is still charged.

Severity Scale

Rating	Meaning
0	Not a usability problem
1	Cosmetic
2	Minor, low priority
3	Major, high priority

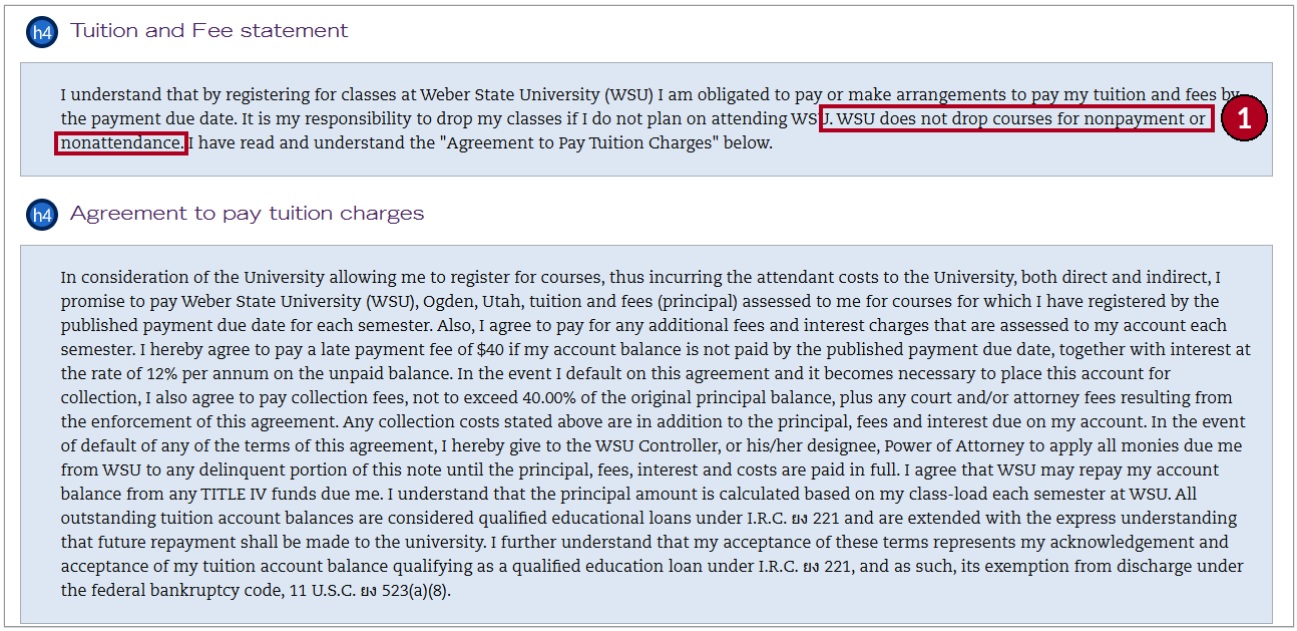
Rating	Meaning
4	Usability catastrophe

Major Issues (Severity 3)

HE01: Add looks like registering, but it isn't (H1 Visibility of system status, H5 Error prevention) Clicking Add only moves a class into the Summary as pending. DevTools showed Add and Submit call different server endpoints, and nothing is registered until the student clicks Submit in the far bottom right corner of the page. A student can close the page believing they're registered when they're not. *Recommendation:* Label the list "Pending, not registered yet," and make the button say exactly what it does, such as "Register for 2 classes."

HE02: The full class error doesn't say the class is full (H9 Help users recognize, diagnose, and recover from errors) When I tried to add a full class, the system showed a vague error that never mentioned the class was full. I couldn't tell whether to retry, choose another section, or join a waitlist. *Recommendation:* "This section is full. Join the waitlist or choose another section."

HE03: The tuition warning is buried and shown only once (H5 Error prevention) Weber State doesn't drop students for nonpayment or nonattendance. That warning appears once, inside several paragraphs of legal text on the Select the Semester page, before the student picks any classes, and isn't repeated near Submit. *Recommendation:* Repeat a short, plain language drop warning right next to the Submit button.



(1) The "WSU does not drop courses" warning, buried in legal text.

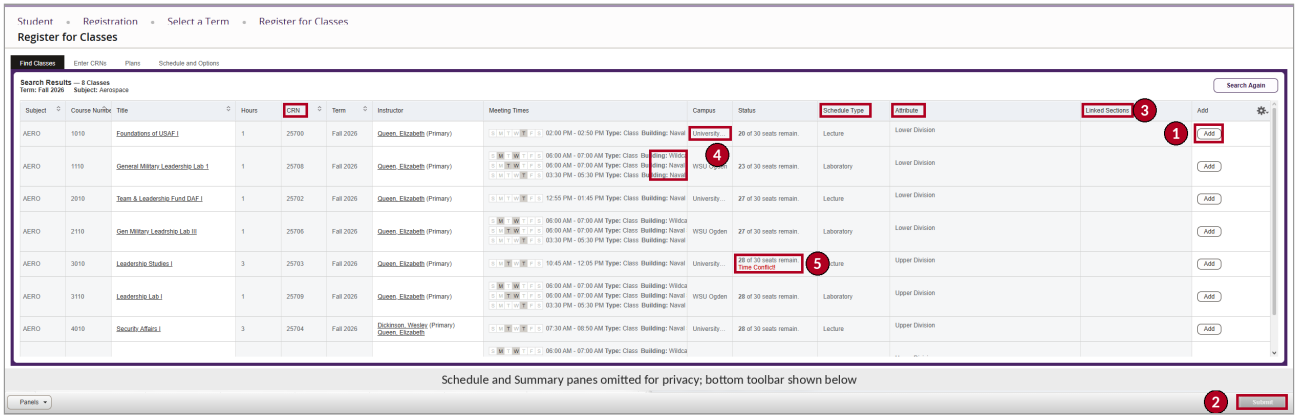
HE12: Zooming in hides content and the Add button (H7 Flexibility and efficiency of use) At 200% zoom the results table header and many values are clipped. At 400%, seat counts can't be read and the Add button is cut off. *Recommendation:* Use a layout that stacks on narrow screens.

HE13: Class times are the hardest thing to read (H8 Aesthetic and minimalist design, H2 Match between system and real world) Meeting days are shown as tiny gray letter boxes with one highlighted, mixed into a cell with Type and Building text that is often cut off. This was found in the user sessions: the novice needed help to find times, and the experienced student called it the hardest task. *Recommendation:* Show days and times in plain text ("Tue/Thu 2:00 to 2:50 PM") in their own column, and add a filter by time of day.

HE14: Prerequisites aren't on the registration page (H6 Recognition rather than recall, H7 Flexibility and efficiency of use) Students must open CatTracks in a separate tab or ask an advisor to know whether they can take a class, then carry that information back. Also found in the user sessions. *Recommendation:* Show prerequisites, and whether the student has met them, on each result.

Minor Issues (Severity 2)

ID	Heuristic	Issue	Recommendation
HE04	H1, H9	Submit is disabled with no explanation of why.	"Add a class to enable registration."
HE05	H8	A waitlist line ("100 of 100 waitlist seats") shows on every section, even ones with open seats.	Show waitlist details only when a section is full.
HE06	H8	Table text is cut off even on a full screen, while large areas of the page sit empty.	Rebalance panes and let cells wrap.
HE07	H2	Jargon like CRN, Linked Sections, Attribute, and Schedule Type is never explained.	Use plain labels or inline definitions.
HE08	H6, H4	Dropping a class is hidden in an Action dropdown.	Use a visible Drop button.
HE09	H10	No format hints on the search form; help is behind a small info icon.	Add hints under each label.
HE10	H6	Students must remember CRNs, pending vs. registered status, and outside approval steps.	Keep pending and registered visibly separate.
HE11	H8	Add is at the top, results at the bottom, and Submit in the far corner, all in 10 to 12 px text.	Confirm next to Add; use 16 px text.



(1) Add, (2) Submit, (3) Jargon columns, (4) Cut off text, (5) Time Conflict warning

What Works

- Green “Registered” labels give clear status after Submit succeeds.
- The weekly calendar shows how classes fit together and makes time conflicts visible.
- A red “Time Conflict!” warning appears right in the search results, before the student clicks Add.
- The breadcrumb shows where the student is in the process.
- The Total Hours line helps students stay under the 20 credit maximum.

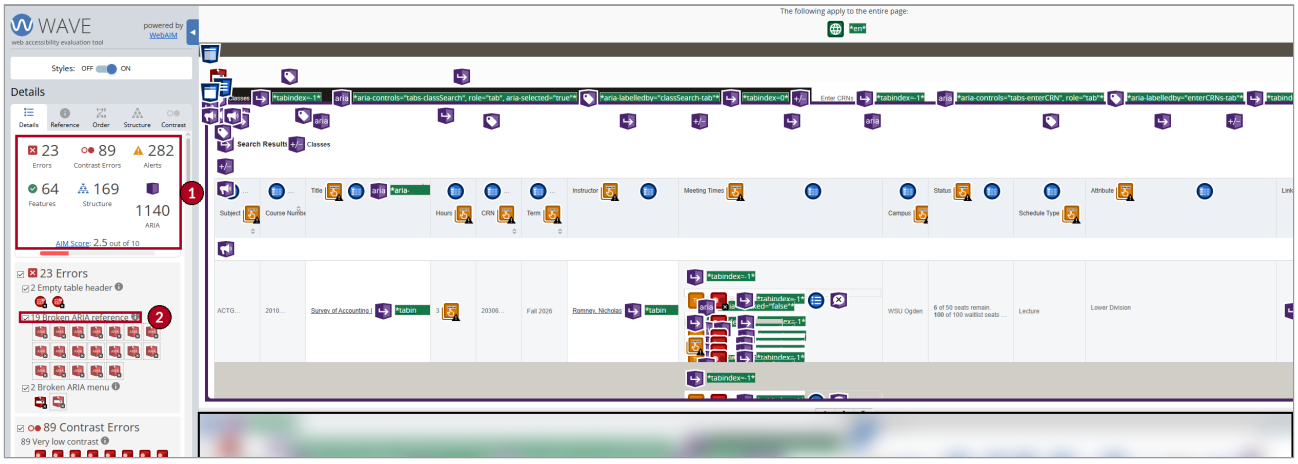
6. Accessibility Audit Findings

> **Can a keyboard only user register for a class?**
>
> **No.** Tab skips every listed class and every Add button.

Automated Scan (WAVE)

Page	Errors	Contrast errors	Alerts	AIM Score
Registration home	4	1	1	8.5
Select the Semester and tuition agreement	0	0	9	9.8
Term selection	3	1	5	8.4
Register for Classes, before searching	3	1	71	8.7
Register for Classes, with search results	**23**	**89**	**282**	**2.5**

The pages leading up to registration score reasonably well. The moment search results load, accessibility collapses, which is exactly where students make their decisions.



(1) WAVE summary with an AIM Score of 2.5 of 10, (2) 19 broken ARIA references

Color Contrast (SC 1.4.3)

Element	Ratio	Result
Active “Schedule” tab label (white on #F7F7F7)	1.07:1	Fail
Primary buttons, light part of gradient	2.85:1	Fail
Primary buttons, middle of gradient	4.42:1	Fail
Gray banner headings on the first two screens	Flagged by WAVE	Fail
Body text (#333333 on white)	12.63:1	Pass

Keyboard and Focus (SC 2.1.1, 2.4.7)

Tabbing through search results skips the listed classes and their Add buttons entirely, so a keyboard user can’t add a class. There is no focus indicator on those buttons, and where focus does appear, it is faint. The disabled Submit button can’t be focused and gives no explanation.

Zoom and Reflow (SC 1.4.4, 1.4.10)

At 200% zoom, the table header and values are clipped. At 400%, seat counts are unreadable and the Add button is cut off. Results grid text is only 10 to 12 px.

Semantic Structure and ARIA (SC 1.3.1, 4.1.2)

The page sets a language and uses `<header>`, a labeled breadcrumb `<nav>`, and `<main>`. However, the three working panes have no headings or labels, the search results contain 19 broken ARIA references and 2 empty table headers, broken ARIA menus appear on nearly every page, and many form labels aren’t connected to their fields.

Findings Summary

ID	WCAG SC	Level	Issue	Severity
A01	2.1.1 Keyboard	A	Add buttons can’t be reached by keyboard	4
A02	2.4.7 Focus Visible	AA	No or faint focus indicator	3
A03	1.4.3 Contrast	AA	Schedule tab label at 1.07:1	3
A05	1.4.10 Reflow, 1.4.4 Resize	AA	Content and Add button clipped when zoomed	3
A08	3.3.3 Error Suggestion	AA	Full class error gives no cause or fix	3
A09	4.1.2, 1.3.1	A	19 broken ARIA references, 2 empty table headers	3
A10	1.4.3 Contrast	AA	89 low contrast items on search results	3
A04	1.4.3 Contrast	AA	Button labels as low as 2.85:1	2
A06	1.3.1, 2.4.6	A / AA	Main panes have no headings or labels	2

ID	WCAG SC	Level	Issue	Severity
A07	3.3.2 Labels or Instructions	A	No format hints on search fields	2
A11	1.4.3 Contrast	AA	Gray banner headings fail contrast	2
A12	1.3.1, 4.1.2	A	Broken ARIA menus, unconnected form labels	2

7. Prioritized Issues

Issues confirmed by more than one method are listed first.

Priority	Issue	Source	Severity	Fix
1	Add looks like registering, but nothing is registered until Submit	Sessions (P1, P2) + Heuristic (HE01, HE04, HE08, HE11)	3	“Pending” label and a button that says “Register for 2 classes”
2	Class times are hard to find and read	Sessions (P1, P2) + Heuristic (HE13, HE06, HE11)	3	Plain text days and times, time of day filter
3	Keyboard users can’t reach Add buttons, with no focus indicator	Accessibility (A01, A02)	4	Real focusable buttons with a thick focus outline
4	Page clipped on phones and when zoomed	Sessions (P1) + Heuristic (HE12) + Accessibility (A05)	3	Responsive layout, 16 px text
5	Prerequisites require CatTracks or an advisor	Sessions (P1, P2) + Heuristic (HE14, HE10)	3	Show prerequisites and status on each result
6	Full class error doesn’t explain the problem or next step	Heuristic (HE02) + Accessibility (A08)	3	Name the cause and next step; announce with `aria-live`
7	Search results page: 19 broken ARIA references, 89 contrast errors	Accessibility (A09, A10)	3	Fix ARIA IDs, raise contrast to 4.5:1
8	Tuition and drop warning buried in legal text	Heuristic (HE03)	3	Plain warning next to Submit
9	Waitlist noise and unexplained jargon in results	Heuristic (HE05, HE07)	2	Plain seat status, plain labels
10	Hidden drop control, no form hints, unlabeled panes	Heuristic (HE08, HE09) + Accessibility (A06, A07)	2	Visible Drop button, hints, labeled sections

8. Post Mortem Reflection

Question 1: How did combining qualitative user observations with an expert heuristic evaluation reveal usability flaws that a developer testing their own code would likely overlook?

A developer testing their own code already knows what every button does. They know Add only stages a class and Submit is what registers it, they know what a CRN is, and they know where the drop control is hidden. Because they built the mental model, they can't experience the gap between what the screen shows and what a student expects. The clearest moment was when my novice participant, trying to pick a section, asked "Where do i see the time for the class." To anyone who built the page, the highlighted day boxes in the Meeting Times column are obvious. To her, the most important fact on the screen was invisible, and she needed help to find it. Watching someone without that knowledge made the gap visible in a way that reading the code never would.

The heuristic evaluation caught a different set of problems: the ones users work around without saying anything. My experienced participant was silent and finished every task without trouble, but she also told me she normally avoids the portal and uses the Schedule Builder and her advisor instead. Her session looked smooth only because she had learned to route around the system. Experienced users blame themselves or quietly adapt, so their sessions can look smooth while the interface is still costing them time. Going screen by screen against the heuristics, and testing with a keyboard, zoom, and WAVE, exposed issues no participant would think to mention, like the Add buttons being unreachable by keyboard.

Each method also confirmed the other. The Add versus Submit split was my top heuristic finding, and the sessions backed it up: my novice participant said that the first time she registered, she thought clicking Add was all she had to do. At the same time, the sessions surfaced two major problems my heuristic review had underrated, unreadable class times and prerequisites that live in a different system. Seeing the same problem from a real user and from a structured review gave me much more confidence that it was a real priority and not just my opinion.

Question 2: Identify one heuristic violation that directly creates a legal or ethical accessibility barrier under WCAG 2.1 AA, and explain how resolving that violation improves the user experience for both disabled users and the general public.

The clearest example is the full class error (HE02), which violates Nielsen's ninth heuristic, help users recognize, diagnose, and recover from errors. When I tried to add a full class, the system showed a vague message that never said the class was full or what to do next. Under WCAG 2.1 AA, this fails SC 3.3.3 Error Suggestion, which requires that when an error is detected and a fix is known, the fix is offered to the user. The system clearly knows the reason, because it rejected the request, but it doesn't share it.

For a screen reader user, this is more than an annoyance. Combined with the missing live region announcements and broken ARIA references on the results page, they may not even notice an error appeared, let alone figure out why. A student with a cognitive disability may not be able to work out the

cause on their own. Because Weber State is a public university covered by the ADA and Section 504, a registration process that some students can't complete independently is a legal risk, and ethically, it means those students depend on someone else to do something every other student does alone.

Fixing it helps everyone. An error that says "This section is full. Join the waitlist or choose another section," announced with an `aria-live` region, tells a screen reader user exactly what happened, but it also saves every student from guessing, retrying, or emailing an advisor. Better yet, showing seat status in plain words before the student clicks prevents the error entirely. That's the pattern I saw across this audit: changes made for accessibility, like clear messages, visible focus, and layouts that reflow when zoomed, end up making the system faster and less stressful for all students.